

Xuhao Luo

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Education

University of Illinois Urbana-Champaign Ph.D. Candidate in Computer Science	Aug. 2021 - Nov. 2026 (Anticipated)
University of California San Diego M.S. in Computer Science	Sep. 2019 - Mar. 2021
University of Science and Technology of China B.S. in Applied Physics	Sep. 2015 - Jun. 2019

Research Publication

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- **Xuhao Luo**, Emaan Atique, Aishwarya Ganesan, Ramnatthan Alagappan, **SkyLog: Reducing Cost and Latency in a Multi-Cloud Shared Log** (In submission to NSDI)
 - **Xuhao Luo**, Shreesha G. Bhat*, Jiyu Hu*(equal contribution), Aishwarya Ganesan, Ramnatthan Alagappan, **LazyLog: A New Shared Log Abstraction for Low-Latency Applications (SOSP 2024) *Best Paper Award, Invited to ACM ToCS***
 - **Xuhao Luo**, Ramnatthan Alagappan, Aishwarya Ganesan, **SplitFT: Fault Tolerance for Disaggregated Datacenters via Remote Memory Logging (EuroSys 2024)**
 - **Xuhao Luo**, Weihai Shen, Shuai Mu, Tianyin Xu, **DepFast: Orchestrating Code of Quorum Systems (USENIX ATC 2022)**
 - Shreesha G. Bhat, Tony Hong, **Xuhao Luo**, Jiyu Hu, Aishwarya Ganesan, Ramnatthan Alagappan, **Low End-to-End Latency atop a Speculative Shared Log with Fix-Ante Ordering (OSDI 2025)**
 - Zhiyuan Guo*, Yizhou Shan*(co-first author), **Xuhao Luo**, Yutong Huang, Yiyang Zhang, **Clio: A Hardware-Software Co-Designed Disaggregated Memory System (ASPLOS 2022)**

Experience

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| Google
<i>Software Engineer Intern, Host: Xiaozhou (Steve) Li, Sr. Staff Engineer</i> | May. 2026 - Aug. 2026
<i>Sunnyvale, CA, USA</i> |
| · Designed and implemented a general framework to extend Bigtable compaction with user-defined functions. Deployed in production and demonstrated its feasibility to replace certain data processing pipeline. | |
| Amazon Web Service
<i>Applied Scientist Intern, Mentor: Prof. George Amvrosiadis, Visiting Scholar</i> | May. 2024 - Aug. 2024
<i>Seattle, WA, USA</i> |
| · Investigated and improved ShardStore (storage system used by AWS S3) reclamation, and built a tool to evaluate different reclamation policy. | |
| Amazon Web Service
<i>Applied Scientist Intern, Mentor: Shen Li, Principle Engineer</i> | May. 2022 - Aug. 2022
<i>Seattle, WA, USA</i> |
| · Improved the performance and reliability of the volume metadata updating workflow for AWS S3 volume metadata cache service with Amazon Quantum Ledger Database (QLDB). | |
| University of Illinois Urbana-Champaign
<i>Research Assistant</i> | May. 2021 - Now
<i>Urbana, IL, USA</i> |
| · <i>SkyLog</i> : Built a shared log abstraction for multicloud applications, with seamless reconfiguration and reduced egress cost. | |
| · <i>LazyLog</i> : Built a new shared log system that offers low-latency append to latency-critical applications by lazily ordering log entries. | |
| · <i>SplitFT</i> : Built a new fault-tolerant approach for storage-centric cloud applications by replicating WALs on remote nodes using RDMA. | |
| · <i>DepFast</i> : Built a framework using C++ coroutine to implement and reason about fail-slow tolerant distributed systems in an easy and effective way. | |
| Microsoft Research
<i>Research Intern</i> | Jun. 2020 - Sep. 2020
<i>Beijing, China</i> |
| · Designed and implemented a task scheduling and dispatching system for distributed machine learning using C++, and a CUDA-based high-performance inter-GPU communication channel for distributed ML. | |
| University of California San Diego
<i>Research Assistant, advised by Prof. Yiyang Zhang</i> | Sep. 2019 - Dec. 2020
<i>La Jolla, CA, USA</i> |
| · Implemented a hardware/software co-designed network stack on both FPGA and host server for disaggregated memory. | |

- Designed and implemented a general 8-bit quantized convolution module on Xilinx Virtex FPGA.
- Developed the TensorFlow C++API for the hardware accelerator using OpenCL. On ResNet-50 CNN, achieved a speedup of 5.17x and a memory usage reduction of 66% compared with the CPU TensorFlow implementation on Xeon E5 2686.

Skills

Language	C/C++, Python, Go, Java, Rust, Haskell, Verilog
Tools/Framework	RDMA, vLLM, SGLang, TensorFlow, Docker, Zookeeper, LLVM

Honors and Awards

• SOSP’24 Best Paper Award	Nov 2024
• SOSP’24 Student Travel Grant	Nov 2024
• EuroSys’24 Student Travel Grant	Apr 2024
• ASPLOS’22 Student Travel Grant	Feb 2022
• USTC Class of 2019 Outstanding Graduates	May 2019

Services

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- EuroSys’26 Shadow PC
 - OSDI’23 Artifact Evaluation Committee
 - USENIX ATC’23 Artifact Evaluation Committee
 - ACM TAAS (Transactions on Autonomous and Adaptive Systems) Invited Reviewer